



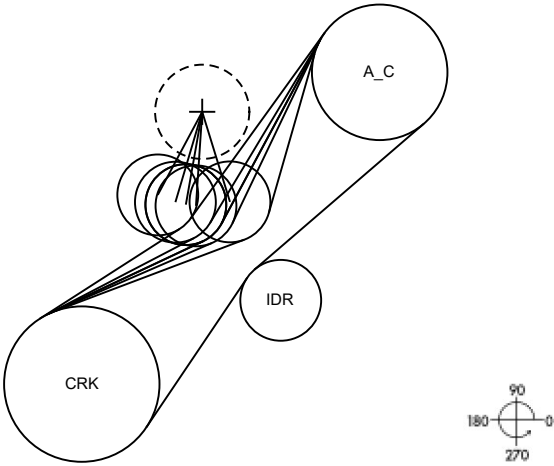
AG0902 BMC Otomotiv
Valeo TM21 AC ø127 Drive
E9843, 22Nm@27°, CCW@115/260
8PK1275HD Fleetranner
IDR@190/80; 08Dec2023

Belt Data, DB Ver. = 2.27.0.0, Belt Rev. = 7/14/2016

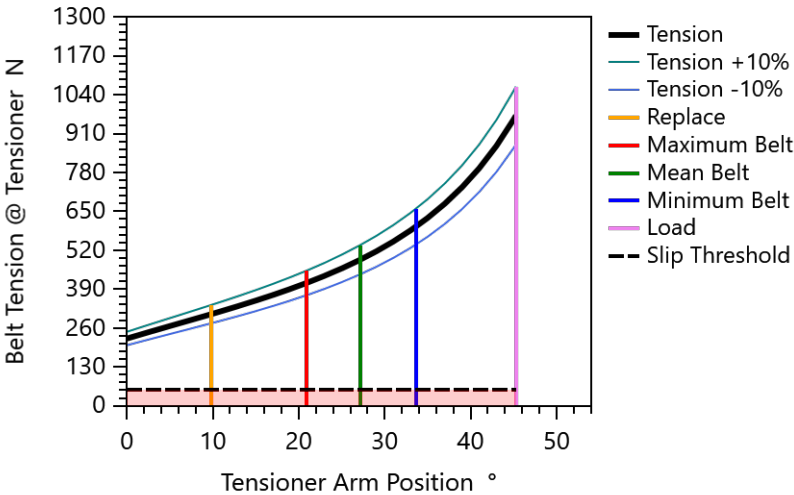
Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Effective Belt Length mm (ISO 9981)	1275

Tensioner Data

Type	Automatic
Design Tension N	488 (110 lbf)



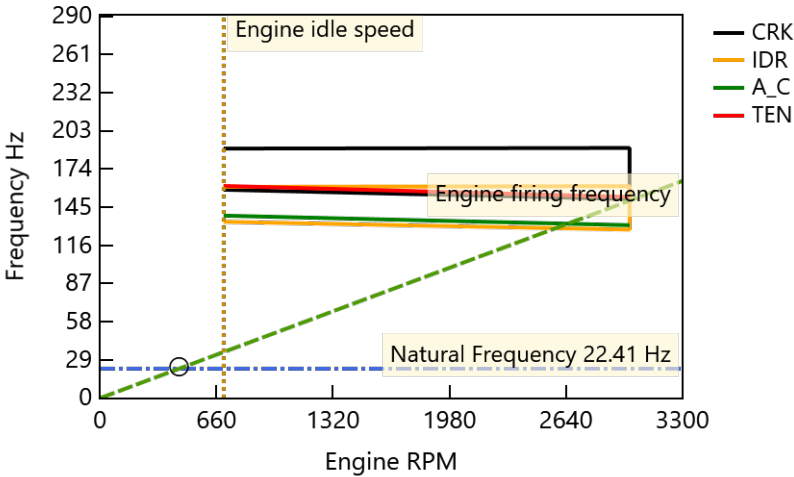
Belt Tension Control



Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr	PK-2_2p-MT3 hours
95	1	32247

Natural Frequency Map



Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °
CRK	2347	1000	807	1251.2	44 / 211
IDR	2347	1000	805	214.3	319 / 15
A_C	2347	1000	491	1275.5	229 / 201
TEN	700	1000	488	307.3	134 / 37

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

Permissible Grooved Pulley

Axial Offset Allowance

For 0.90° Belt Entry Angle

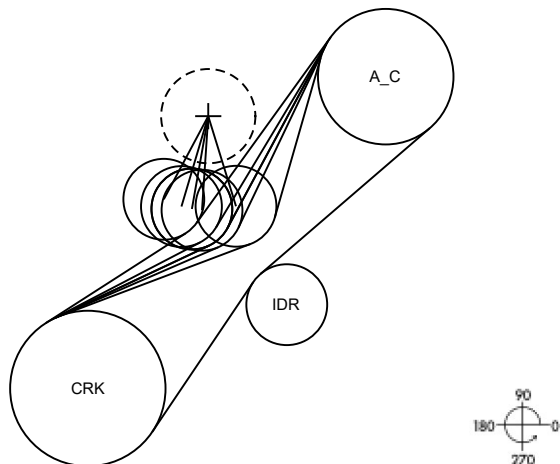
Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
A_C->CRK	IDR (0.33)	2.76	66	34
CRK->A_C	TEN (1.20)	1.0	5	95



AG0902 BMC Otomotiv
Valeo TM21 AC ø127 Drive
E9843, 22Nm@27°, CCW@115/260
8PK1275HD Fleetranner
IDR@190/80; 08Dec2023

Layout Data mm

	X	Y	Flat	Pitch	Effective	DOB
CRK	0.00	0.00		148.40	146.00	146.99
IDR	190.00	80.00	75.00	77.20	79.60	
A_C	284.50	297.70		129.40	127.00	127.99
TEN	99.37	171.37	75.00	77.20	79.60	



Belt Data, DB Ver. = 2.27.0.0, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Stretch and Wear Allowance % of Length	0.70
Belt Length Tolerance (+/-) mm	5.00

Tensioner Data

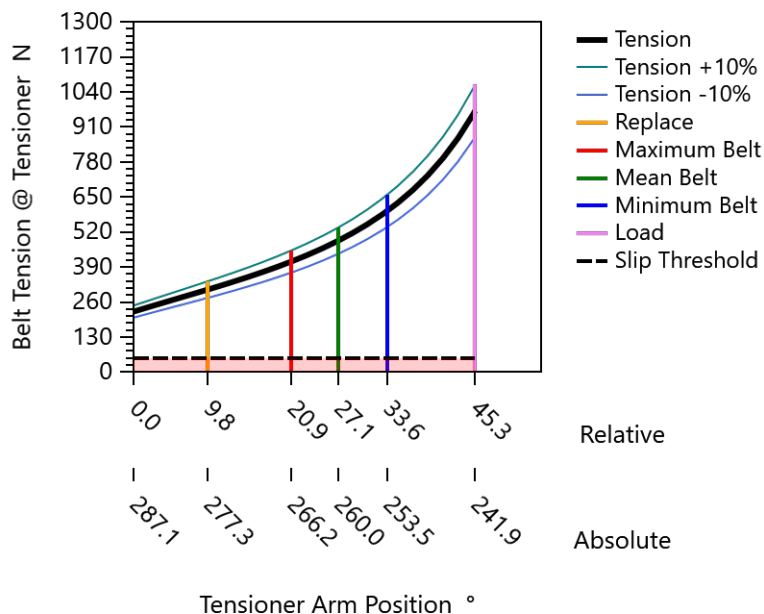
Type	Automatic
Design Tension N	488 (110 lbf)
Pivot Point {X, Y Coordinates} mm	115.00 , 260.00
Arm Length mm	90.0
Spring Mean Load Nm	22.15
Spring Pre-Load Nm	9.13
Spring Rate Nm/deg	0.480

RESULTS

Belt Drive System Geometry

	Span Length mm	Wrap Angle °	Speed Ratio (Ref. Engine)
CRK	172.6	210.8	1.000
IDR	213.7	15.3	1.922
A_C	198.9	201.0	1.147
TEN	162.8	36.6	1.922

Belt Tension Control





AG0902 BMC Otomotiv
Valeo TM21 AC ø127 Drive
E9843, 22Nm@27°, CCW@115/260
8PK1275HD Fleetrunner
IDR@190/80; 08Dec2023

Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio mm/°

0.792

Position Description	Belt-Drive Length					
	Free Arm	Replace	Max	Mean	Min	Load
Arm Position °	287.1	277.3	266.2	260.0	253.5	241.9
Idler X Coordinate mm	141.5	126.4	109.0	99.4	89.4	72.6
Idler Y Coordinate mm	174.0	170.7	170.2	171.4	173.7	180.6
Hubload to Arm Angle °	30.2	38.1	47.7	53.5	59.8	71.6
Hubload Direction °	137.3	135.4	133.9	133.5	133.2	133.4
Hubload N	201.6	249.3	287.8	306.3	325.1	361.4
Tension N	224.8	306.5	410.8	488.1	599.2	970.3
Idler Wrap Angle °	53.3	48.0	41.0	36.6	31.5	21.5
Effective Drive Length mm	1296.7	1289.3	1280.4	1275.4	1270.4	1262.8
Required Eff. Belt Length mm	1296.0	1288.6	1279.7	1274.7	1269.7	1262.1

Belt Take-up

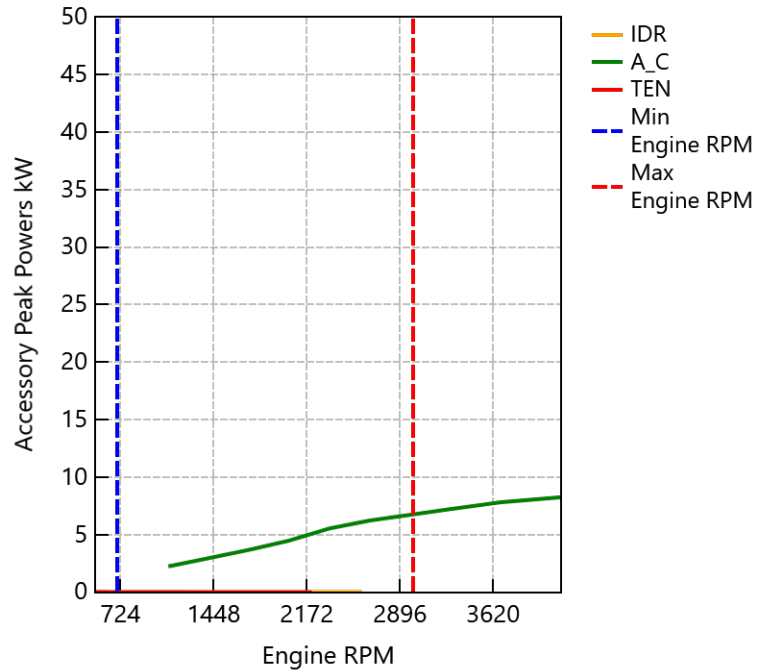




AG0902 BMC Otomotiv
Valeo TM21 AC ø127 Drive
E9843, 22Nm@27°, CCW@115/260
8PK1275HD Fleetranner
IDR@190/80; 08Dec2023

Design Tension N	488 (110 [lbf])
Engine Acceleration RPM/s	1000
Engine Deceleration RPM/s	1000

Accessory Peak Powers kW



Accessory Data

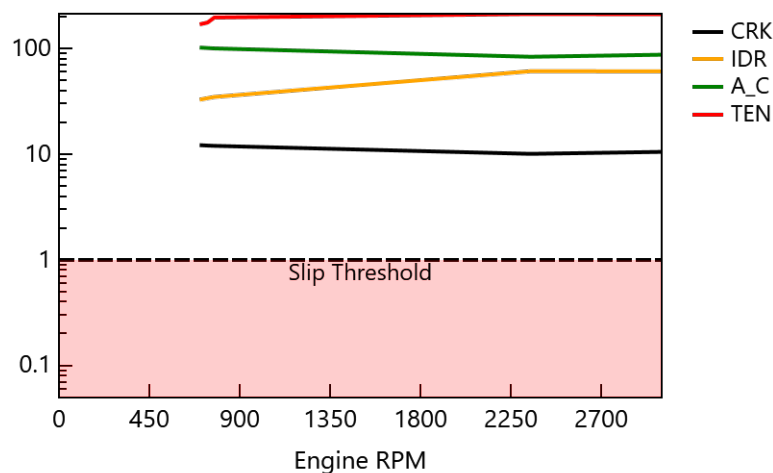
	File Name - if used	Moment of Inertia kg m ²
IDR	IDR.cmp	0.0002
A_C	Valeo-TM21-7_9kW_A_C.cmp	0.0060
TEN	TEN.cmp	0.0004

RESULTS

Belt Slip Sensitivity

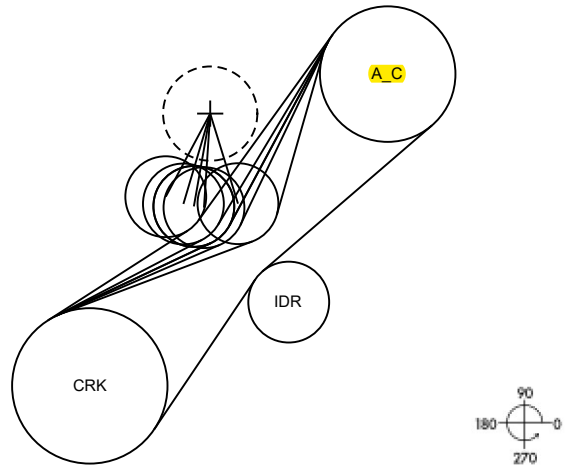
Pulley	Most Critical Load Condition					Required Increase Tension or Wrap Angle	
	Acceler. RPM/s	Load Combination %	A_C			N	°
CRK	1000	100					
IDR	1000	10					
A_C	1000	100					
TEN	1000	100					

Belt Slip Safety Factor





AG0902 BMC Otomotiv
Valeo TM21 AC ø127 Drive
E9843, 22Nm@27°, CCW@115/260
8PK1275HD Fleetrunner
IDR@190/80; 08Dec2023



Load Conditions kW for DC 95%

%	Eng. RPM	V, km/hr	T, °C	CRK	IDR	A_C	TEN
35.00	700	1	70	2.51	0.80	1.70	0.01
45.00	1200	1	70	5.01	2.00	3.00	0.01
19.00	2000	1	70	8.21	3.60	4.60	0.01
1.00	3000	1	70	11.21	4.00	7.20	0.01

Mean Hubloads N for DC 95%

%	Eng RPM	CRK	IDR	A_C	TEN
35.00	700	1391	275	1272	307
45.00	1200	1466	324	1280	307
19.00	2000	1457	332	1253	306
1.00	3000	1410	290	1266	306

Mean Tensions N for DC 95%

%	Eng RPM	CRK	IDR	A_C	TEN
35.00	700	950	802	490	488
45.00	1200	1025	811	489	488
19.00	2000	1016	785	489	488
1.00	3000	969	797	488	488